

1 Highlights

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- 6 • Land-derived Landsat/Sentinel-2 harmonization is invalid over dark
7 oligotrophic water
- 8 • Sensors stay 99.8% separable after harmonization; intercepts exceed
9 signal 70-fold
- 10 • Campaign, not station, is the blocking unit in campaign-based moni-
11 toring programmes
- 12 • Honest Secchi retrieval: $R^2=0.574$, RMSE 1.86 m over 812 real match-
13 ups
- 14 • Conformal intervals are calibrated marginally but under-cover in the
15 clearest water

Harmonization and validation limits of satellite Secchi retrieval in Lake Titicaca

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Abstract

Monitoring water transparency in high-altitude oligotrophic lakes is critical for freshwater management, yet it remains constrained by the sparsity of field data. While multi-mission satellite archives (e.g., harmonized Sentinel-2 and Landsat 8/9) and machine learning offer a synoptic solution, current practices often import assumptions from turbid-water literature that fail in clear-water, campaign-based settings. We assemble 1002 satellite–field match-ups against twelve years of limnological monitoring in Lake Titicaca to address these failures and define a defensible retrieval protocol. First, we demonstrate that standard land-derived harmonization coefficients (e.g., Roy et al., 2016) are invalid over dark water, as their intercepts exceed the native Landsat signal by factors of 1.4 to 69.8, leaving sensors 99.8% separable; we propose a classifier two-sample test to catch this failure. Second, we show that station-blocked cross-validation, the standard remedy for spatial autocorrelation, leaves the dominant dependence in campaign-based sampling untouched: under station-blocked folds a null model that knows only the campaign date still explains $R^2 = 0.305$ of the variance. We introduce a null-model diagnostic that identifies the blocking unit — here the cam-

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38 paign — rather than assuming it. Finally, under this strict design, we train
 39 a Random Forest regressor to retrieve Secchi disk depth ($R^2 = 0.574$, RMSE
 40 1.86 m) and attach split-conformal prediction intervals. We reveal that while
 41 marginally calibrated (89.4% coverage at 90% nominal), conformal intervals
 42 structurally under-cover in the clearest water (71.3%). Our findings provide
 43 a rigorous blueprint for evaluating machine-learning water quality products
 44 in large, oligotrophic lakes.

45 *Keywords:* Lake Titicaca, Secchi disk depth, Sentinel-2, Landsat,
 46 cross-sensor harmonization, conformal prediction

47 1. Introduction

48 Large lakes are sentinels of environmental change, but continuous moni-
 49 toring of remote, high-altitude systems is constrained by sparse field infras-
 50 tructure (Williamson et al., 2009). Lake Titicaca (3,810 m a.s.l.), the largest
 51 tropical lake in the world by volume, is a critical transboundary freshwater
 52 reserve whose water quality is documented only by infrequent field cam-
 53 paigns (OEFA, 2016; Siguayro Mamani et al., 2022), even as eutrophication
 54 advances in its near-shore bays (Heredia et al., 2022; Lanza et al., 2024).
 55 Satellite remote sensing offers a synoptic solution (Gholizadeh et al., 2016;
 56 Topp et al., 2020). Specifically, water transparency—quantified by the Secchi
 57 disk depth (Z_{SD})—is an integrative indicator of water clarity strongly linked
 58 to light attenuation (Lee et al., 2015). Because Sentinel-2 MSI and Landsat
 59 8/9 OLI provide complementary revisit times, their joint use as a virtual
 60 constellation coupled with machine learning is now the standard practice for
 61 mapping water quality (Pahlevan et al., 2020; Cao et al., 2022).

62 Although these virtual constellations work well in turbid, productive
 63 “Case-2” waters, they fail to transfer reliably to clear, oligotrophic high-
 64 altitude lakes due to two unresolved methodological assumptions. First,
 65 the cross-sensor harmonization coefficients required to merge Landsat and
 66 Sentinel-2 (e.g., Roy et al., 2016) were fitted on terrestrial targets. Because
 67 terrestrial surfaces are orders of magnitude brighter than clear water, their
 68 additive intercepts overwhelm the over-water signal. While near-infrared
 69 bias over water has been noted (Pahlevan et al., 2019), the magnitude of
 70 this mismatch and its effect on machine-learning retrievals remain unquan-
 71 tified. Second, to prevent spatial autocorrelation from inflating apparent
 72 accuracy, previous methods routinely block cross-validation by monitoring
 73 station (Roberts et al., 2017). However, they assume rather than test that
 74 station identity is the primary source of data leakage. In campaign-based
 75 monitoring programmes, where an entire lake is sampled within a few days
 76 under identical conditions, station-blocking may retain severe leakage and
 77 falsely inflate performance metrics.

78 In this paper, we propose a rigorous validation and retrieval pipeline to
 79 overcome these challenges, using Lake Titicaca as our testing ground. Our
 80 innovation is replacing assumed methodological defaults with data-driven
 81 diagnostics. Specifically, we assemble 1002 real satellite–field match-ups and
 82 apply a classifier two-sample test (adversarial validation) (Lopez-Paz and
 83 Oquab, 2017) to diagnose harmonization failure. Furthermore, we introduce
 84 a null-model diagnostic to identify the true blocking unit for cross-validation.
 85 Finally, we train a Random Forest regressor under the corrected design and
 86 attach split-conformal prediction intervals (Angelopoulos and Bates, 2023)

87 to quantify per-prediction uncertainty.

88 In contrast to previous water-quality mapping efforts (Ayala Izurieta
89 et al., 2023; Maligaya et al., 2024), our study makes three explicit contri-
90 butions:

- 91 **1. Quantification of Harmonization Limits:** We show that land-
92 derived harmonization intercepts exceed the native Landsat signal by
93 up to 69.8-fold over clear water. Consequently, a classifier can identify
94 the source sensor with 99.8% accuracy from harmonized bands, proving
95 that the model learns a sensor label rather than unified optics.
- 96 **2. Null-Model Validation Design:** We demonstrate that under station-
97 blocked folds a null model predicting only the campaign-date mean still
98 achieves $R^2 = 0.305$, so station blocking leaves the dominant, temporal
99 dependence intact. Evaluating null models under each candidate block-
100 ing unit identifies the campaign as the unit that carries the dependence
101 in this class of data.
- 102 **3. Defensible Retrieval with Conformal Uncertainty:** Under this
103 strict design, our model achieves an honest R^2 of 0.574 (RMSE 1.86 m).
104 We further reveal that while split-conformal intervals are marginally
105 calibrated (89.4% coverage), they structurally under-cover extreme val-
106 ues in the clearest water, highlighting a critical limitation for opera-
107 tional monitoring.

108 2. Materials and methods

109 2.1. Study area

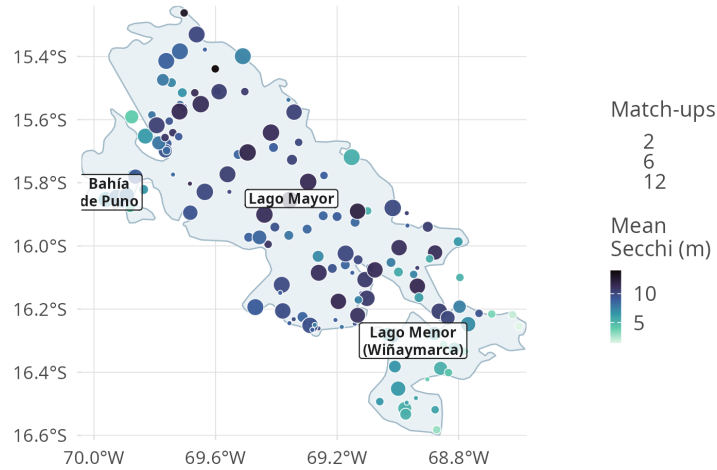
110 Lake Titicaca comprises three limnologically distinct zones (Fig. 1): the
111 large, deep, oligotrophic Lago Mayor; the shallower Lago Menor (Wiñay-
112 marca) to the south-east; and Bahía de Puno, which receives municipal and
113 agricultural inputs from the city of Puno and is the most eutrophic sector.
114 Measured transparency spans 1.5 m in near-shore turbid water to 16.5 m in
115 the open oligotrophic basin.

116 2.2. In-situ data

117 We used 881 field measurements of Secchi transparency, with co-measured
118 chlorophyll-*a*, suspended solids, temperature, pH, dissolved oxygen and nu-
119 trients, from the limnological monitoring campaigns of IMARPE (Instituto
120 del Mar del Perú) on Lake Titicaca, spanning 2011–2024 across 160 sta-
121 tion identifiers distributed over the whole lake. Six of those identifiers are
122 a 2017 naming variant of a station already recorded under another label,
123 so the records describe 154 distinct stations; we retain the identifiers as
124 issued, which makes station-blocked validation marginally more permissive
125 and therefore does not favour the argument of Section 2.6. Of the 881 mea-
126 surements, 734 include a valid Secchi reading. The July 2019 campaign (71
127 stations) is published by Siguayro Mamani and Franco (2022); the remaining
128 years derive from the same IMARPE programme. As a provenance check, our
129 2019 zone-level means reproduce the published values almost exactly (Lago
130 Mayor conductivity 1513.4 vs. 1513.5 $\mu\text{S cm}^{-1}$; Lago Menor chlorophyll-*a*
131 1.52 vs. 1.52 mg m^{-3}).

(a) Limnological monitoring network (IMARPE)

IMARPE/ALT stations with satellite-field match-ups (2013–2024), coloured by mean measured Secchi depth. The trophic gradient is visible in the field data alone



(b) Sampling effort is very uneven

Each point is a station. 143 stations, but most were visited in very few campaigns, limiting the power of any station-based analysis

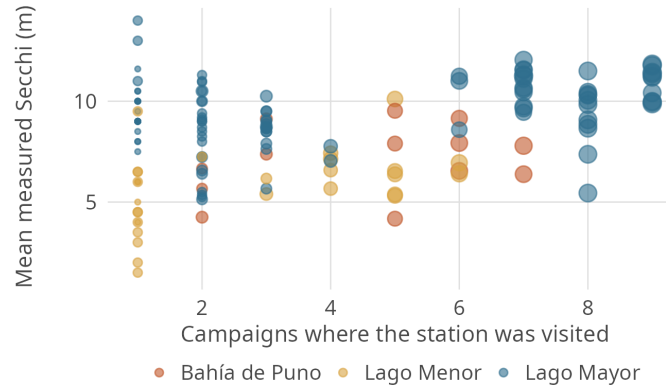


Figure 1: Study area and monitoring network. (a) IMARPE/ALT stations with a satellite–field match-up, coloured by mean measured Secchi depth over the Natural Earth lake boundary; the trophic gradient is evident in the field data alone. (b) Sampling effort per station: most stations were visited in few campaigns, which limits the power of any station-level analysis.

132 Three properties of this record constrain everything that follows, and we
133 state them here rather than in the limitations (Fig. A.5).

134 First, the monitoring programme is itself almost entirely a dry-season
135 programme. The 2011–2024 record contains 13 campaigns with Secchi read-
136 ings, of which exactly one — December 2012, 72 measurements — falls in the
137 austral wet season (December–March). That campaign predates both sen-
138 sors used here: Landsat 8 became operational in May 2013 and Sentinel-2A
139 in June 2015. Consequently every match-up in our dataset falls between July
140 and October, and this is a property of the field programme rather than of
141 our processing: no choice of match-up window or cloud threshold can recover
142 wet-season pairs that have no contemporaneous imagery. We return to what
143 this means for operational use in Section 4.3.

144 Second, sampling is episodic: campaigns exist for 2013–2024 with no
145 campaigns in 2020, 2021, 2023.

146 Third, the reference measurement is coarse: 73 distinct Secchi values
147 occur in 812 match-ups, because disk readings are recorded to the nearest
148 half metre.

149 No synthetic or spectrally derived labels were used at any stage.

150 *2.3. Satellite data and cross-sensor harmonization*

151 Sentinel-2 MSI (harmonized surface reflectance) and Landsat 8/9 OLI
152 (Collection 2, Level-2; Vermote et al., 2016) were accessed through Google
153 Earth Engine (Gorelick et al., 2017). Clouds were masked with the Sentinel-
154 2 Scene Classification Layer plus cloud probability, and with the Landsat
155 QA_PIXEL bitmask (dilated cloud, cloud and shadow). The cirrus flag was
156 not applied, as it is near-permanently set at 3810 m and would mask the

157 entire archive. We retained the six bands common to both sensors (blue,
158 green, red, NIR, SWIR1, SWIR2).

159 Landsat surface reflectance was first harmonized to Sentinel-2 equivalent
160 reflectance with the per-band linear coefficients of Roy et al. (2016), of the
161 form $\rho_{\text{MSI}} = a + b \rho_{\text{OLI}}$. Section 3.1 evaluates whether this transform is valid
162 over the water of Lake Titicaca and derives an alternative. For that evalu-
163 ation we identified every in-situ event observed by both sensors within the
164 match-up window — 270 paired station–date observations — and regressed
165 Sentinel-2 on harmonized Landsat reflectance band by band.

166 2.4. *Satellite–field match-ups*

167 For every in-situ measurement we extracted the cloud-masked median
168 reflectance within a 30 m buffer of the station coordinate, from imagery
169 acquired within ± 10 days. The 30 m buffer matches the coarser Landsat pixel
170 and averages residual geolocation error and sensor noise over several pixels
171 while remaining in open water, away from shoreline mixing. The ± 10 -day
172 window balances the scarce cloud-free image availability at 3810 m against
173 the slow temporal change of transparency in this large, deep lake, and is
174 consistent with windows used in operational Z_{SD} retrieval (Kloiber et al.,
175 2002; Zhang et al., 2022).

176 This yielded 1002 match-ups (340 Sentinel-2, 662 Landsat), of which 812
177 have a Secchi value, across 143 stations and 120 campaign dates. Because
178 both sensors can observe the same field event, 272 match-ups are cross-sensor
179 pairs of a single in-situ measurement; collapsing them leaves 540 independent
180 events, and Section 3.2 verifies that accuracy is unchanged, confirming the
181 result is not an artefact of pseudo-replication.

182 2.5. Features and model

183 Twelve sensor-agnostic features were computed from the six common
184 bands: the bands themselves plus NDWI, NDTI and the ratios blue/green,
185 green/blue, red/green and blue/red. We trained a Random Forest regres-
186 sor (Breiman, 2001). Reflectance was scaled with a robust (median/IQR)
187 scaler fitted only on the training partition of each fold. All models were
188 implemented with scikit-learn (Pedregosa et al., 2011).

189 Model choice was settled empirically rather than assumed. We com-
190 pared Random Forest, Extremely Randomized Trees, XGBoost (Chen and
191 Guestrin, 2016) and LightGBM, each with hyperparameters optimized by
192 Optuna (Akiba et al., 2019) inside *nested* cross-validation — tuning on an
193 inner campaign-blocked split of each outer training partition, so no outer fold
194 ever informs its own hyperparameters — and benchmarked the resulting gain
195 against the spread across random seeds. Section 3.4 reports the outcome.

196 2.6. Validation design and null models

197 Rather than a single split, we evaluate a hierarchy of increasingly de-
198 manding designs (Roberts et al., 2017): random K -fold; GroupKFold blocked
199 by station; GroupKFold blocked by *campaign date*; GroupKFold blocked by
200 year; a temporal hold-out; and leave-one-zone-out, in which an entire trophic
201 zone is withheld.

202 Crucially, we do not assume which blocking unit matters. We benchmark
203 each design against null models that predict, respectively, the training-set
204 mean of the zone, of the station and of the campaign date. A blocking unit
205 is doing real work if, and only if, the corresponding null model collapses

under it. This diagnostic, rather than convention, determines our primary design.

Performance is reported as out-of-fold R^2 , RMSE and MAE, and benchmarked against a classical log-linear blue/green band-ratio algorithm in the standard form $\ln Z_{SD} \sim \ln(\rho_{\text{blue}}/\rho_{\text{green}})$ (Kloiber et al., 2002), with predictions bounded to the physically observable range, and against a multiband linear model.

2.7. Uncertainty and interpretation

We attach a per-prediction interval using split-conformal prediction (Angelopoulos and Bates, 2023): within each fold, a calibration subset of the training partition yields the $(1 - \alpha)$ quantile of absolute residuals, defining a symmetric interval with finite-sample marginal coverage guarantees (Romano et al., 2019). Because a single nominal level can be met by chance, we report empirical coverage at 50, 80, 90 and 95%, and also coverage conditional on measured transparency, since marginal calibration does not imply conditional calibration.

The model is interpreted with SHAP (Lundberg and Lee, 2017; Lundberg et al., 2020), computed *out of fold* — each observation explained by the model that did not see it — rather than on a model fitted to the whole dataset, which inflates the apparent importance of memorized features. Trends in annual median in-situ transparency were tested with Mann–Kendall and Sen’s slope (Mann, 1945; Sen, 1968) on the full 2011–2024 in-situ record, not on the match-up subset.

229 3. Results

230 3.1. Land-derived harmonization is invalid over oligotrophic water

231 The Roy et al. (2016) transforms carry additive intercepts of 0.0116–
232 0.0448 in reflectance units. Over Lake Titicaca the native Landsat over-water
233 signal is far smaller than this (Table A.3, Fig. 2a). In the near-infrared the
234 median native reflectance is effectively zero, so the intercept exceeds the
235 signal by a factor of 69.8; in SWIR1 the factor is 23.2, in red 10.6, and even
236 in blue — the brightest water band — 2.5. The harmonized Landsat bands
237 are consequently dominated by a constant inherited from a land calibration:
238 their medians exceed the corresponding Sentinel-2 medians by 1.6-fold in
239 green and 15.4-fold in the near-infrared.

240 A harmonization that succeeded would make the two sensors statistically
241 indistinguishable over the same water. They are not. A Random Forest clas-
242 sifier trained to identify the source sensor from the harmonized features alone
243 reaches 99.8% accuracy against a 66.3% majority-class baseline (Fig. 2b); re-
244 stricting it to the visible bands gives 97.9%, and to scale-invariant band ratios
245 99.5%. This is a classifier two-sample test (Friedman, 2003; Lopez-Paz and
246 Oquab, 2017): if two samples come from the same distribution, no classifier
247 can separate them above chance. Sensor identity therefore remains fully en-
248 coded in the harmonized reflectance, and any model trained on the pooled
249 archive can read it. The test is not new, but to our knowledge it has not
250 been used as a harmonization acceptance check; it costs one model fit and it
251 is decisive.

252 Refitting locally on the 270 station–date events observed by both sensors
253 gives water-specific coefficients (Table A.4). These restore the spectral sig-

254 nature almost exactly — the median Landsat/Sentinel-2 ratio falls from 15.4
 255 to about 1.0 across all six bands (Fig. 2d) — but per-observation agreement
 256 remains uneven (Fig. 2c). Green recovers usable agreement ($r = 0.71$), and
 257 NIR and SWIR reach $r \approx 0.6$. Blue does not: $r = 0.26$. This is the most
 258 consequential result of the section, because SHAP attributes 72.6% of the
 259 retrieval to green/blue bands and ratios (Section 3.7). The band on which
 260 the retrieval most depends is the band on which the two sensors least agree.

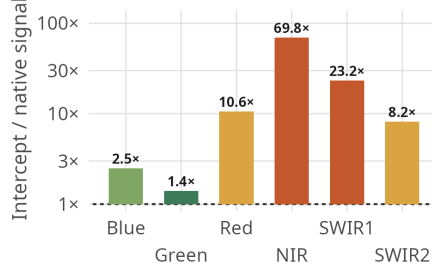
261 *3.2. The campaign, not the station, is the blocking unit*

262 Table 2 reports the retrieval under each validation design (Fig. A.6), and
 263 Table 1 the three null models under each blocking design. Blocking by station
 264 changes nothing relative to a random split (both $R^2 = 0.596$), which under
 265 the usual reading would be taken as evidence of no spatial leakage. The
 266 null models show why that reading is wrong. Each blocking unit collapses
 267 its own null by construction; what matters is what survives. Under station-
 268 blocked folds a null that predicts only the training-set mean of the campaign
 269 date still attains $R^2 = 0.305$: the temporal structure passes through station
 270 blocking untouched. The converse does not hold in the same way. Under
 271 campaign-date-blocked folds the station-mean null retains $R^2 = 0.208$, but
 272 the zone-mean null retains 0.287 under that design and between 0.234 and
 273 0.276 under every other, so what station identity carries is essentially the
 274 trophic gradient — a property of the lake that a retrieval should learn, not
 275 leakage. Dependence in this dataset lives between campaigns, not between
 276 stations, because a single campaign samples the whole lake within a few days
 277 under one set of atmospheric and limnological conditions.

278 Under campaign-date-blocked validation — our primary design — the

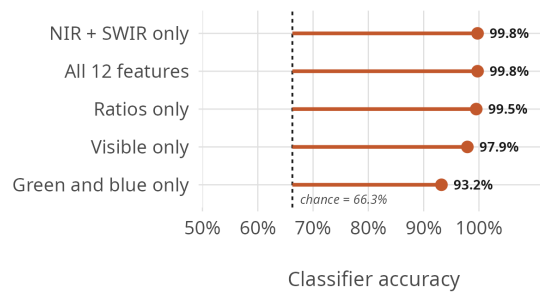
(a) Land intercept exceeds the signal

Roy et al. (2016) coefficients, fitted over vegetation and soil, vs native Landsat reflectance over the lake water. The dashed line marks intercept = signal



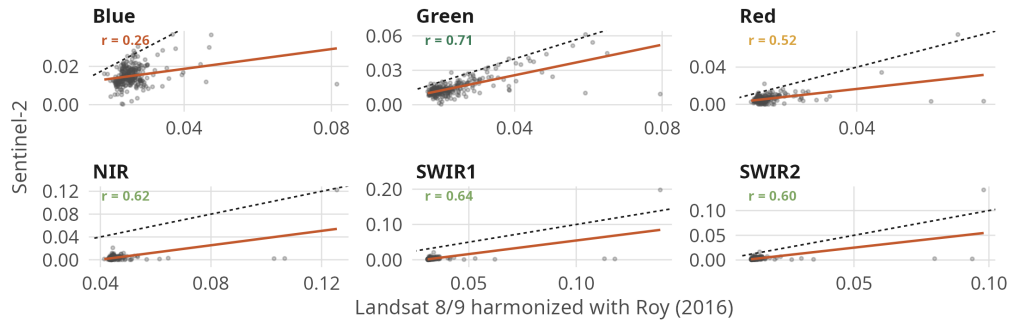
(b) Sensors remain distinguishable

Accuracy of a Random Forest predicting the source sensor from already harmonized reflectance. If harmonization worked, it would be at chance



(c) Blue is the band that agrees least between sensors

Sentinel-2 reflectance vs Roy-harmonized Landsat in the 270 events observed by both sensors at the same station and date. The dashed line is 1:1



(d) Local recalibration over water does harmonize bands

Median reflectance over water by sensor. Labels are the Landsat / Sentinel-2 ratio: it goes from 15.4x in NIR to ~1.0x in every band

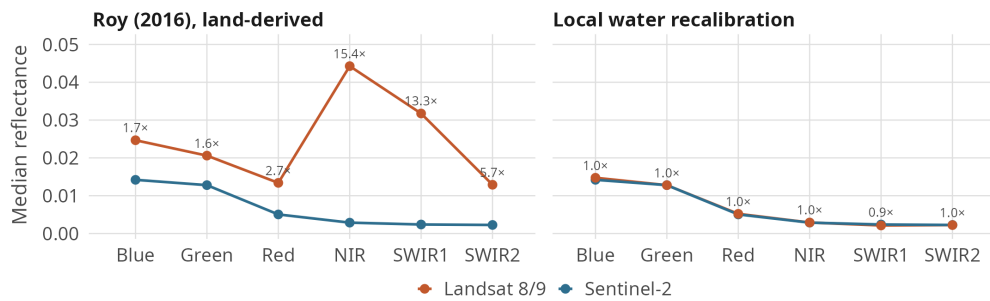


Figure 2: Why land-derived harmonization fails over clear water. (a) The Roy intercept relative to the native over-water Landsat signal, by band. (b) Accuracy of a classifier predicting the source sensor from harmonized features; under a successful harmonization it would sit at chance. (c) Sentinel-2 versus harmonized Landsat reflectance on the 270 paired station-date events; blue agrees worst. (d) Median spectral signature before and after local water recalibration; labels are the Landsat/Sentinel-2 ratio.

Table 1: Null models under each blocking design (out-of-fold R^2 , $n = 812$). Each null predicts only the training-set mean of one grouping, and each blocking unit collapses its own null by construction (diagonal). What survives off the diagonal is the dependence that design leaves intact.

Null model	Folds blocked by		
	Station	Campaign date	Year ^a
Zone-mean	0.276	0.287	0.234
Station-mean	-0.008	0.208	0.222
Campaign-date-mean	0.305	-0.012	-0.032

^a The programme ran one field campaign per year, so year blocking isolates whole campaigns.

279 Random Forest attains $R^2 = 0.574$ and RMSE 1.86 m over 812 match-ups,
 280 essentially unbiased (bias -0.02 m). The campaign date is the finest unit at
 281 which the campaign structure can be expressed, but not the coarsest: the 9
 282 field campaigns in the match-up record each span 7–27 days (median 13 sam-
 283 pling dates), so date-blocked folds can still place two days of one campaign
 284 on opposite sides of the split. Blocking by year isolates whole campaigns,
 285 since the programme ran one per year, and gives $R^2 = 0.495$; the difference of
 286 0.079 in R^2 bounds the residual within-campaign dependence. We retain date
 287 blocking as the primary design because nine year-groups would leave five-fold
 288 GroupKFold with fewer than two campaigns per fold and too few calibration
 289 residuals for the conformal analysis, and we report the year-blocked figure
 290 alongside it as the stricter bound. Collapsing cross-sensor pseudo-replicates
 291 to 540 independent events leaves accuracy unchanged ($R^2 = 0.571$).

Table 2: Retrieval under each validation design (Random Forest, $n = 812$ unless noted).
Null models are in Table 1.

	Setting	R^2	RMSE (m)
Blocking unit	Random K -fold	0.596	1.81
	By station	0.596	1.81
	By campaign date	0.574	1.86
	By year	0.495	2.02
	One record per event ($n = 540$)	0.571	1.77
Zone extrapolation	Leave out Lago Menor ($n = 133$)	0.247	1.90
	Leave out Bahía de Puno ($n = 88$)	0.036	2.11
	Leave out Lago Mayor ($n = 591$)	-0.047	2.52
Temporal	Train ≤ 2019 , test 2022 & 2024 ($n = 226$)	0.507	2.63

292 3.3. Retrieval accuracy and benchmarks

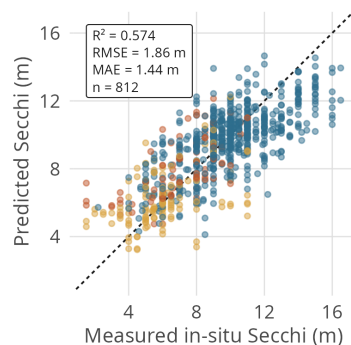
293 Against this design the classical blue/green band ratio attains $R^2 = 0.097$
294 (RMSE 2.71 m) and a multiband linear model $R^2 = 0.323$ (RMSE 2.34 m)
295 (Table A.5, Fig. 3). We report one methodological caution alongside these
296 figures. An unbounded implementation of the same band-ratio algorithm —
297 log-transformed and back-transformed without constraining predictions to
298 the observable range — produces a single prediction of 143.5 m against an
299 observed maximum of 16.5 m, which alone drives R^2 to -2.458 and RMSE to
300 5.29 m (Fig. 3d). Benchmarks of classical algorithms reported without phys-
301 ical bounds should be treated with caution: a single unconstrained extrapo-
302 lation manufactures an arbitrarily large advantage for the machine-learning
303 model.

304 3.4. Model family and tuning do not change the picture

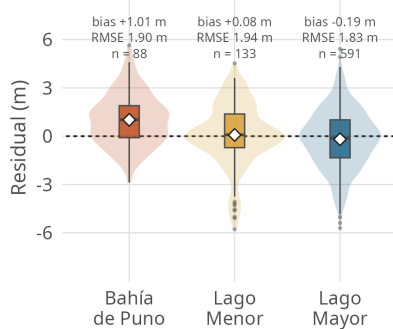
305 We compared four learner families, each tuned by Optuna inside nested
306 cross-validation, against target transformations and reduced feature sets (Ta-
307 ble A.6). The best configuration — an equal-weight ensemble of the four
308 tuned models — reached $R^2 = 0.583$, an improvement of 0.009 over the
309 untuned Random Forest. That gain does exceed the spread across random
310 seeds (0.003 in R^2), so it is not noise; but it corresponds to 0.02 m of RMSE
311 on a lake whose reference measurement is recorded to the nearest half metre,
312 and it changes no conclusion in this article. Neither log-transforming the
313 target nor tuning individual learners helped more than that, and restricting
314 the feature set cost accuracy.

315 We therefore retain the untuned Random Forest as the reported model:
316 it is simpler, faster and easier to reproduce, and reporting a tuned ensemble

(a) Predicted vs measured
Random Forest, campaign-blocked
validation. Dashed line is 1:1



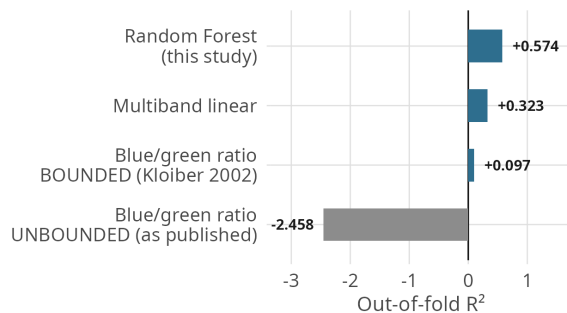
(b) Bias concentrates in the bay
Residuals (predicted - measured) by zone.
Diamond is the mean. Bahía de Puno, the
most eutrophic zone, +1.0 m



• Bahía de Puno • Lago Menor • Lago Mayor

(c) The baseline was misimplemented

Same validation design for all.
The published baseline undid the log without
bounds; a single 143 m prediction ruined its R^2



(d) One impossible value ruins R^2

Classical ratio on log scale.
The 143 m point is the one that produced
the $R^2 = -2.46$

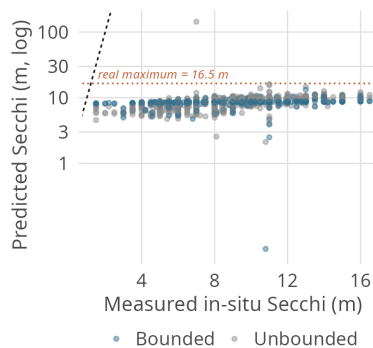


Figure 3: Retrieval under the primary design. (a) Out-of-fold predicted versus measured Secchi depth. (b) Residuals by trophic zone; the retrieval overestimates in eutrophic Bahía de Puno. (c) Benchmark against classical and linear baselines. (d) The unbounded classical implementation in log scale: one impossible prediction destroys the metric.

317 for a 0.009 gain would overstate the precision of the product. The ceiling
 318 here is set by the data — a dry-season-only record, a reference measurement
 319 quantized at half a metre, and a target whose variance is dominated by a
 320 three-level trophic gradient — not by the choice of learner.

321 3.5. *Where the retrieval stops working*

322 Withholding an entire trophic zone collapses the retrieval (Table 2): R^2
 323 falls to 0.247 for Lago Menor, 0.036 for Bahía de Puno and -0.047 for Lago
 324 Mayor. The model interpolates within optical regimes it has seen and does
 325 not extrapolate to a water type absent from training. On the temporal hold-
 326 out (train ≤ 2019 , test 2022 and 2024) it retains $R^2 = 0.507$, though 57 of the
 327 69 test stations also appear in training (82.6%), so this measures temporal
 328 rather than fully independent generalization.

329 Error is unevenly distributed (Table A.7). The retrieval carries a positive
 330 bias of 1.01 m in Bahía de Puno, the most eutrophic and most vulnerable
 331 zone, because the clear-water majority pulls predictions upward.

332 Two features of the harmonized archive should be read together with these
 333 numbers. Removing the two bands most corrupted by the land-derived in-
 334 tercepts lowers R^2 only from 0.574 to 0.541; using scale-invariant ratios alone
 335 gives 0.468; and applying our water-specific recalibration gives 0.518. Train-
 336 ing on each sensor separately gives 0.563 (Sentinel-2, $n = 274$) and 0.549
 337 (Landsat, $n = 538$). The pooled model’s small advantage over these vari-
 338 ants is consistent with it exploiting sensor identity as an additional covariate
 339 rather than with genuine cross-sensor synergy.

340 3.6. Calibrated per-prediction uncertainty

341 Split-conformal intervals are well calibrated *marginally* at every nominal
342 level tested: empirical coverage is 50.9% at 50% nominal, 79.1% at 80%,
343 89.4% at 90% and 94.3% at 95%, a maximum calibration error of 0.94 per-
344 centage points (Fig. A.7a).

345 They are not calibrated *conditionally*, and this matters more for opera-
346 tional use than the marginal result. Coverage falls to 71.3% in the clearest
347 water ($Z_{SD} > 12.5$ m) and 79.5% in the most turbid ($Z_{SD} < 5$ m), while
348 exceeding the nominal level in the middle of the range (Fig. A.7c). The
349 mechanism is structural rather than incidental: split conformal derives a sin-
350 gle global quantile of absolute residuals, so the interval has constant width
351 (6.12 m at 90%) everywhere and cannot adapt to the local difficulty of the re-
352 trieval. It therefore buys its marginal guarantee by over-covering in the well-
353 sampled middle of the range and under-covering at both extremes — which
354 are precisely where management decisions are made. A locally adaptive vari-
355 ant such as conformalized quantile regression (Romano et al., 2019) would
356 be required for reliable conditional coverage, and we recommend reporting
357 coverage conditional on the predicted value as standard practice whenever
358 conformal intervals accompany an operational product.

359 For a lake whose transparency spans 1.5–16.5 m, a ± 3 m interval is in-
360 formative for zone-scale contrasts and multi-metre change, and is not infor-
361 mative for sub-metre differences.

362 3.7. Interpretation and retrievability

363 Out-of-fold SHAP attributes 72.6% of the retrieval to green and blue
364 bands and their ratios, 14.4% to NIR and SWIR, and the remainder to

other indices (Fig. 4a). Green/blue alone accounts for 37%. The dependence is monotone and of the sign expected from water-clarity optics (Fig. 4b): a higher green-to-blue ratio pushes the prediction toward less transparent water. The retrieval is therefore grounded in optics rather than location memorization — while resting on the band pair with the worst cross-sensor agreement.

With real labels and the same campaign-blocked design, no other variable is retrievable (Fig. 4c): chlorophyll-*a* reaches $R^2 = 0.012$, suspended solids 0.056, in-situ water temperature 0.113, and Landsat thermal surface temperature 0.020 ($r = 0.14$, $n = 662$). Their in-situ ranges — chlorophyll-*a* 0–4.6 mg m⁻³, suspended solids mostly below 10 mg L⁻¹ — lie below the optical and thermal detection limits of these sensors. We subject the negative claim to the same design as the positive one, and report it to caution against applying chlorophyll algorithms (Gitelson et al., 2009; Mishra and Mishra, 2012) or thermal retrievals (Ruiz-Verdú et al., 2016) in clear high-altitude water without local validation.

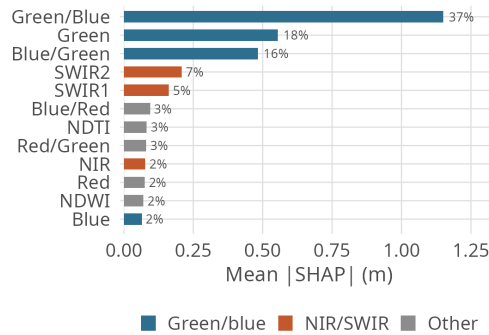
3.8. Temporal patterns and the fragility of trend claims

Mann–Kendall tests on annual median in-situ transparency over the full 2011–2024 record show no statistically significant trend in any zone (Bahía de Puno: Sen slope 0.16 m yr⁻¹, $p = 0.241$; Lago Mayor: 0.25 m yr⁻¹, $p = 0.138$; Fig. A.8a).

That conclusion is fragile in a way worth reporting explicitly. Restricting the series to 2013 onwards — the natural choice if trends were computed on the satellite match-up subset, whose record begins with Landsat 8 — turns both Bahía de Puno ($p = 0.017$) and Lago Mayor (Sen +0.42 m yr⁻¹,

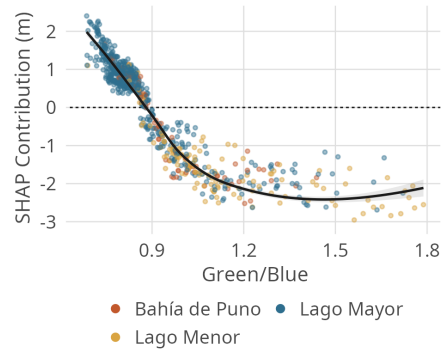
(a) Green and blue dominate attribution

Mean |SHAP| computed OUT-OF-FOLD: each observation is explained by the model that did not see it. Labels are the share of the total



(b) How Green/Blue acts

SHAP value vs. feature value. Greener water relative to blue pushes the prediction toward less transparent water. Trimmed to the 1-99th percentile (18 extreme points out)



(c) Only transparency is retrievable in this water

Random Forest with the same honest design (campaign blocking) for each variable. The negative claim is subject to the same rigor as the positive one

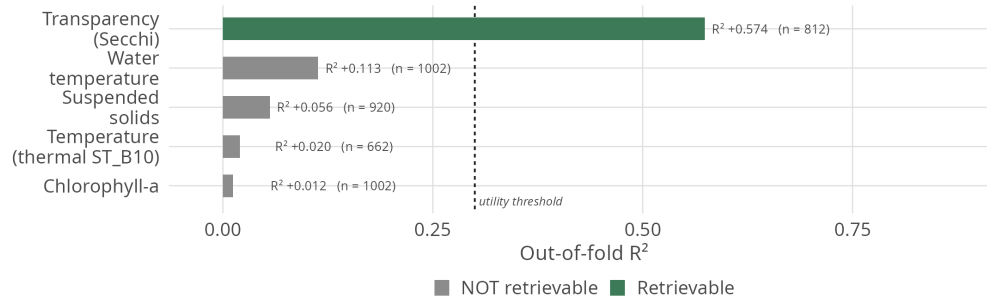


Figure 4: Interpretation and operational envelope. (a) Out-of-fold SHAP importance; green and blue dominate. (b) SHAP dependence on the leading feature, by trophic zone. (c) Retrievability of each water-quality variable under the same campaign-blocked design; only transparency clears a useful threshold.

390 $p = 0.012$) significantly positive. Across start years from 2011 to 2014, 5 of
 391 12 zone–start-year combinations change verdict (Fig. A.8b). The year 2013
 392 has the lowest annual medians on record, so beginning there manufactures
 393 an upward trend. With 9–11 annual values and three gap years the test has
 394 low power, and the correct reading is absence of evidence, not evidence of
 395 stability. We recommend that trend analyses in campaign-based lake records
 396 report this sensitivity as a matter of course.

397 **4. Discussion**

398 *4.1. Implications for harmonized multi-mission archives*

399 The virtual-constellation approach (Claverie et al., 2018; Pahlevan et al.,
 400 2019) has become the default for multi-decadal lake monitoring, and the Roy
 401 et al. (2016) coefficients are its most reused transform. Our results place
 402 a boundary on where that default is safe. Those coefficients were fitted on
 403 terrestrial targets and carry additive intercepts appropriate to surfaces one
 404 to two orders of magnitude brighter than clear water. Over Lake Titicaca the
 405 intercept exceeds the native over-water signal by a factor of 69.8 in the near-
 406 infrared and by 2.5 even in blue, and the two sensors remain 99.8% separable
 407 after the transform. That water reflectance from these products is unreliable
 408 is already documented (Pahlevan et al., 2019; Masek et al., 2021); what our
 409 paired observations add is the magnitude — the problem is not confined
 410 to the near-infrared, and it is large enough to make the harmonized bands
 411 function as sensor labels for a machine-learning model — and the finding that
 412 local recalibration repairs the aggregate spectral signature without repairing
 413 per-observation agreement in blue.

414 The practical consequence is direct. A machine-learning model trained
415 on a pooled, land-harmonized archive can achieve apparently good accuracy
416 while having learned two sensor-specific mappings joined by a readable sensor
417 label. That is not a harmonized retrieval, and it will not behave as one when
418 applied to a scene from either sensor alone. The diagnostic is cheap and we
419 recommend it as routine: train a classifier to predict the source sensor from
420 the harmonized features and report its accuracy against the majority-class
421 baseline. If the sensors remain separable, the harmonization has not done
422 its job. For clear oligotrophic lakes we further recommend deriving water-
423 specific coefficients from same-day paired observations, reporting per-band
424 agreement, and — where blue-band agreement is as poor as we find here —
425 reporting single-sensor retrievals alongside any pooled product.

426 *4.2. Implications for validation design*

427 Blocking by station is the standard remedy for spatial autocorrelation
428 in this literature, and in our data it changes nothing: random and station-
429 blocked folds give identical accuracy. Read conventionally, that would be
430 reassuring. The null models show it is not. Under station-blocked folds
431 a null that knows only the campaign date still reaches $R^2 = 0.305$ — as
432 much as the zone-mean null reaches under any design — so the temporal
433 structure passes through station blocking untouched. In a campaign-based
434 monitoring programme the whole lake is sampled within a few days under
435 shared atmospheric and limnological conditions, so observations cluster by
436 campaign at least as strongly as by location, and only a design that blocks
437 on the campaign removes that clustering.

438 The general point is that the blocking unit should be diagnosed, not

439 assumed. Reporting null models under each candidate design costs little
 440 and identifies the unit that carries the dependence. This applies well beyond
 441 Lake Titicaca: most large-lake monitoring programmes sample in campaigns,
 442 so the same mismatch between the conventional blocking unit and the real
 443 dependence structure is likely to be widespread.

444 For our retrieval, the honest figure is $R^2 = 0.574$ (RMSE 1.86 m) under
 445 campaign-blocked validation, against 0.287 for a zone-mean null: the model
 446 roughly doubles the skill available from knowing only which basin a sample
 447 came from. That is a real but modest contribution, and stating it that way
 448 is more useful to a manager than an unqualified R^2 .

449 *4.3. Operational envelope for Lake Titicaca*

450 Our R^2 of 0.574 sits below the range typically published for Z_{SD} retrieval
 451 ($R^2 \approx 0.6$ – 0.9 ; Zhang et al., 2022; Lee et al., 2015), obtained here under
 452 a predominantly clear oligotrophic regime and a stricter validation design,
 453 and our RMSE of 1.86 m spans roughly 12% of the observed Z_{SD} range. It
 454 is also approaching the granularity of the reference measurement itself: disk
 455 readings are recorded to the nearest half metre, giving only 73 distinct values
 456 across 812 match-ups.

457 Four constraints bound operational use. The retrieval carries a 1.01 m
 458 positive bias in Bahía de Puno — the most eutrophic and most vulnerable
 459 zone — so mapped values there must be read with their intervals and con-
 460 firmed by field sampling. Leave-one-zone-out results show the model cannot
 461 extrapolate to an unsampled optical regime; integration must be incremen-
 462 tal, each new campaign both validating the current map and extending the
 463 training set. The retrieval is calibrated and validated for the austral dry sea-

464 son only, and the reason is worth stating precisely because it determines the
465 remedy. It is not that wet-season imagery is unusable at 3810 m, nor that our
466 match-up criteria excluded it: it is that the field programme has produced
467 a single wet-season campaign in fourteen years, and that campaign predates
468 both satellites. Wet-season blooms in Bahía de Puno — the phenomenon a
469 transparency product would most usefully warn about — are therefore en-
470 tirely unrepresented, and the model should not be applied between December
471 and March. Finally, because conformal coverage degrades at both extremes
472 of the transparency range, the nominal 90% interval should be treated as
473 optimistic in the clearest and most turbid water.

474 This converts the limitation into a specific, actionable recommendation
475 for the monitoring authorities. Extending the protocol to the wet season does
476 not require a better algorithm; it requires field data that do not yet exist. A
477 single additional wet-season campaign per year, timed to any cloud-free win-
478 dow and covering the three trophic zones, would within a few years supply
479 enough independent campaign-blocks to calibrate and validate a wet-season
480 retrieval under the same design used here. Because the binding constraint is
481 sampling rather than sensing, this is among the cheapest available improve-
482 ments to Lake Titicaca’s water-quality monitoring capability.

483 Within those bounds the product is useful to the regional monitoring
484 authorities (IMARPE, ALT, OEFA, and the binational management frame-
485 work). A calibrated ± 3 m interval resolves zone-scale contrasts and multi-
486 metre change, which is the scale at which clarity loss in Bahía de Puno
487 would be actionable, and a single transboundary product under one cali-
488 bration supports comparable assessment across the Peru–Bolivia boundary

where coordinated field monitoring is logistically difficult. It does not resolve sub-metre change and should not be presented as doing so.

4.4. Limitations

Beyond the constraints already stated, three limitations bound this work. The in-situ record is episodic, with no campaigns in 2020, 2021, 2023 and only one wet-season campaign in fourteen years, so the temporal hold-out rests on two test years, the trend analysis on 9–11 annual values, and the seasonal envelope on none. Transparency above approximately 15 m saturates the optical signal. And this is a single lake: the two diagnostics we propose are general, but the water-specific coefficients in Table A.4 are local to Titicaca and must be refitted elsewhere.

5. Conclusions

Using 1002 satellite–field match-ups over twelve years, we show that two practices imported from the turbid-water remote-sensing literature fail in a clear high-altitude large lake. Cross-sensor harmonization coefficients derived over land leave Sentinel-2 and Landsat 99.8% separable over oligotrophic water, because their additive intercepts exceed the native over-water signal by up to a factor of 69.8; and validation blocked by monitoring station leaves the dominant dependence — between campaigns — entirely intact.

Correcting both, a Random Forest retrieves Secchi depth with $R^2 = 0.574$ and RMSE 1.86 m, roughly double the skill of a zone-mean null, with conformal prediction intervals that are calibrated marginally but under-cover at the extremes of the transparency range. Skill collapses under extrapolation to an

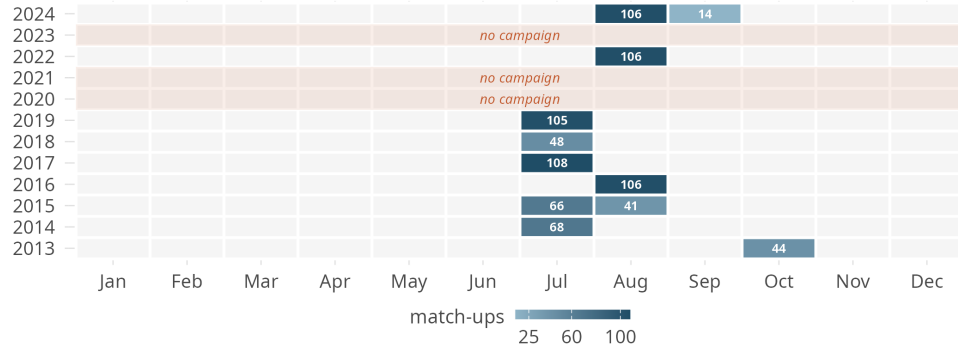
512 unsampled trophic zone; chlorophyll-*a*, suspended solids and thermal tem-
513 perature are not retrievable at all; and no transparency trend is significant
514 once the full in-situ record is used rather than the satellite-era subset.

515 We therefore recommend two routine checks for multi-mission lake water-
516 quality products, neither of which requires new methodology: report the
517 accuracy of a classifier two-sample test on the harmonized features, and re-
518 port null models under each candidate validation blocking unit. Both cost
519 one model fit, and both distinguish genuine optical retrieval from sensor la-
520 belling and sampling design. For Lake Titicaca the resulting product is a
521 dry-season, zone-scale transparency indicator with honest ± 3 m intervals —
522 less than has sometimes been claimed for such lakes, and more than the
523 region has had.

524 **Appendix A. Supplementary Figures and Tables**

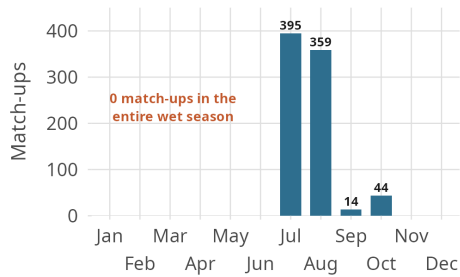
(a) Three years with no campaigns and no data outside the dry season

Number of satellite-field match-ups by year and month. There is no data for 2020, 2021 or 2023, so the 2022-2024 hold-out only contains 2022 and 2024



(b) Only dry season data exists

Bahía de Puno blooms occur during the wet season, which the dataset does not cover at all



■ Wet season (Dec-Mar) ■ Transition ■ Dry season (Jul-Oct)

(c) The trophic gradient dominates Secchi

Only 73 distinct values in 812 match-ups: readings are rounded to half a metre, so the RMSE of 1.86 m borders the granularity of the reference measurement itself

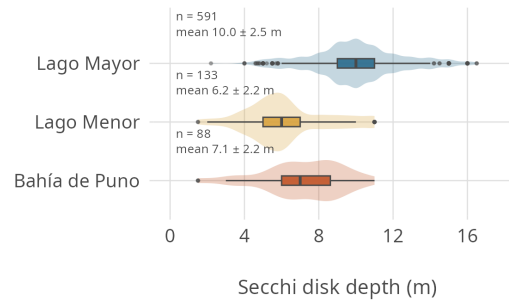
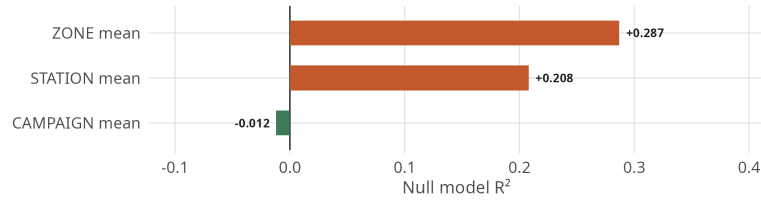


Figure A.5: What the dataset actually contains. (a) Match-ups by year and month; three years have no campaign at all, so a hold-out described as “2022–2024” contains only 2022 and 2024. (b) Seasonal coverage: zero match-ups in the wet season, when blooms occur in Bahía de Puno. (c) Secchi distribution by trophic zone; the reference measurement is quantized at roughly half a metre.

(a) Only campaign blocking neutralizes its own null model

R^2 of models predicting only the group mean, with folds blocked by campaign.

A blocking unit is effective if and only if its own null model collapses to zero under it



(b) Station blocking changes nothing; extrapolation breaks everything

Out-of-fold R^2 of the Random Forest; the large point is the primary design.

The fact that station blocking matches with random does not prove absence of leakage: it proves that it retains nothing

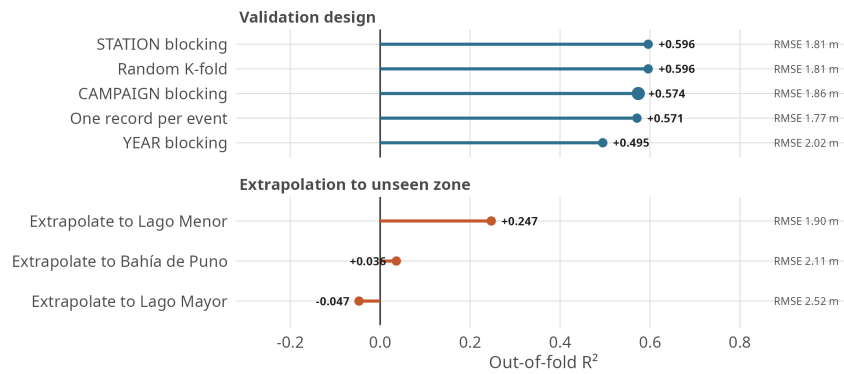
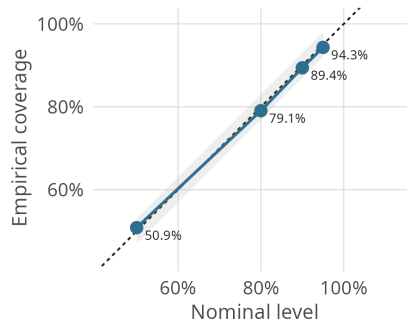
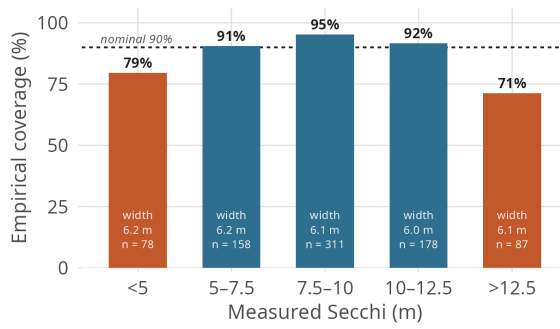


Figure A.6: Diagnosing the blocking unit. (a) Null models evaluated under campaign-date-blocked folds; the same three nulls under every blocking design are in Table 1. (b) Out-of-fold R^2 under each validation design and under extrapolation to an unseen trophic zone. That station blocking matches the random split is not evidence of no leakage; it is evidence that station blocking retains nothing.

(a) Calibrated, on average
Out-of-fold empirical coverage vs nominal. Gray band is ± 3 points



(c) But not at the ends of the range
Split-conformal uses a single global quantile, so the width is constant (~ 6.1 m) and the interval cannot adapt: it undercovers in the clearest water



(b) The 90% interval vs the measurement

Match-ups ordered by measured transparency. The line is the prediction, the band is the 90% conformal interval. 86 out of 812 points fall outside

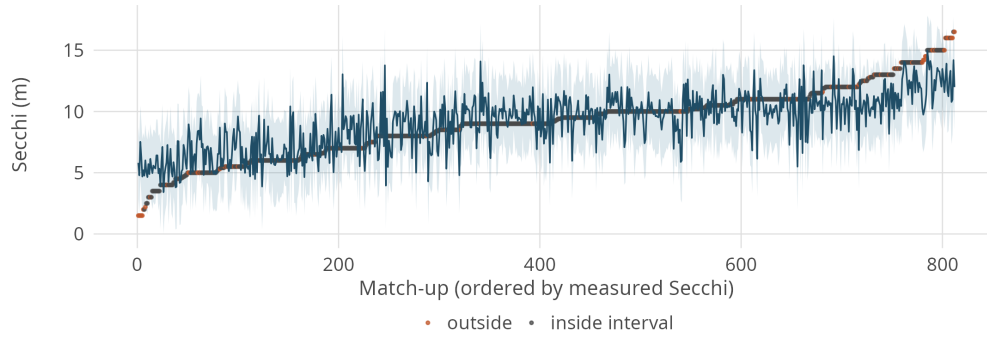
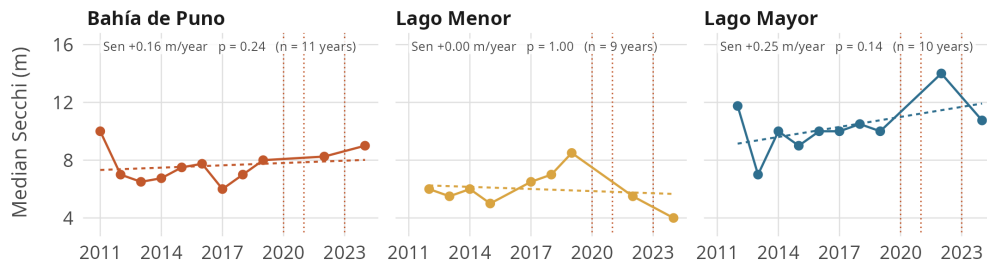


Figure A.7: Conformal prediction intervals under campaign-blocked validation. (a) Empirical versus nominal coverage: marginally calibrated across the 50–95% range. (b) The 90% interval against the measured value. (c) Coverage conditional on measured transparency. Because split conformal uses a single global residual quantile, the interval width is constant (≈ 6.12 m) and cannot adapt, so the marginal guarantee is bought by over-covering in the middle of the range and under-covering at both extremes.

(a) No zone shows significant trend in the full record

Annual median transparency in-situ (2011–2024, 734 readings). The vertical dotted lines mark the years with no campaign: 2020, 2021 and 2023



(b) The verdict is set by the start year, not the lake

Same Mann-Kendall test, same data, different start year. Starting in 2013 — the year with the lowest medians in the record — turns two zones into "significantly more transparent"

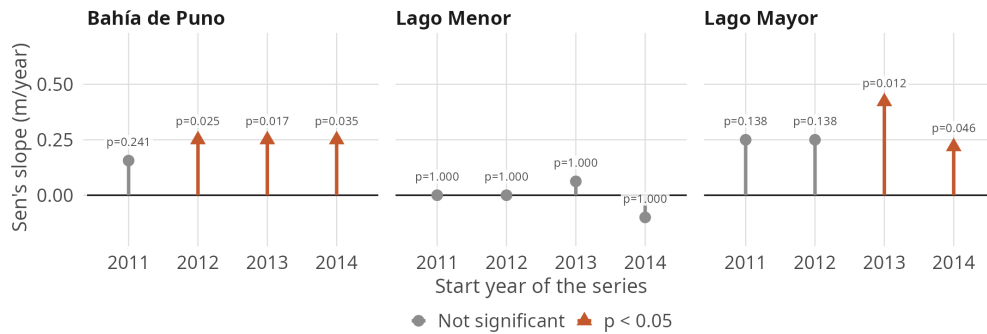


Figure A.8: Transparency trends and their fragility. (a) Annual median in-situ transparency by zone over the full 2011–2024 record, with Mann–Kendall/Sen statistics; no trend is significant. (b) The same test with different start years: beginning the series in 2013, the lowest-median year on record, turns two zones significantly positive.

Table A.3: Effect of the Roy et al. (2016) OLI→MSI transform over Lake Titicaca water. Native OLI is the Collection-2 surface reflectance recovered by inverting the transform. Reflectance is dimensionless.

Band	Roy intercept a	Native OLI median	Intercept / signal	Harmonized LS median	Harmonized / S2 median
Blue	0.0183	+0.0072	2.5	0.0247	1.74
Green	0.0123	+0.0089	1.4	0.0206	1.61
Red	0.0123	+0.0013	10.6	0.0134	2.66
NIR	0.0448	−0.0006	69.8	0.0443	15.42
SWIR1	0.0306	+0.0013	23.2	0.0317	13.34
SWIR2	0.0116	+0.0013	8.2	0.0129	5.70

Table A.4: Water-specific LS→S2 recalibration fitted on 270 paired station–date observations, of the form $\rho_{S2} = a + b \rho_{LS, Roy}$.

Band	a	b	Pearson r	Agreement
Blue	+0.0084	0.258	0.26	unusable
Green	−0.0009	0.665	0.71	good
Red	−0.0005	0.427	0.52	weak
NIR	−0.0250	0.631	0.62	acceptable
SWIR1	−0.0223	0.770	0.64	acceptable
SWIR2	−0.0057	0.615	0.60	acceptable

Table A.5: Head-to-head benchmark, campaign-blocked GroupKFold, $n = 812$.

Algorithm	R^2	RMSE	MAE	Bias
Classical blue/green, unbounded ^a	−2.458	5.29	1.91	−0.19
Classical blue/green, bounded ^b	0.097	2.71	2.13	−0.46
Multiband linear	0.323	2.34	1.80	−0.01
Random Forest (this study)	0.574	1.86	1.44	−0.02

RMSE, MAE and bias in metres.

^a back-transformed without bounding predictions; see text.

^b standard form of Kloiber et al. (2002).

Table A.6: Model selection under nested cross-validation (Optuna inside an inner campaign-blocked split of each outer training partition). Selected configurations; the full grid is in the repository.

Model	Features	R^2	RMSE (m)
Ensemble of the four tuned models	all 12	0.583	1.84
LightGBM, tuned	all 12	0.581	1.84
XGBoost, tuned	all 12	0.577	1.85
Random Forest, tuned	all 12	0.576	1.85
Random Forest, untuned (reported)	all 12	0.574	1.86
Extremely Randomized Trees, untuned	all 12	0.573	1.86
XGBoost, tuned, $\log Z_{SD}$	all 12	0.561	1.89
LightGBM, tuned	visible only	0.544	1.92
LightGBM, tuned	ratios only	0.488	2.04
Seed-to-seed spread of the untuned Random Forest: R^2 range 0.003 over five seeds.			

Table A.7: Out-of-fold error by trophic zone, campaign-blocked validation.

Zone	R^2	RMSE (m)	Bias (m)	n
Bahía de Puno	0.221	1.90	+1.01	88
Lago Menor	0.211	1.94	+0.08	133
Lago Mayor	0.448	1.83	−0.19	591

525 **CRedit authorship contribution statement**

526 **Dina Maribel Yana-Yucra:** Conceptualization, Data curation, For-
527 mal analysis, Investigation, Methodology, Project administration, Resources,
528 Software, Supervision, Validation, Visualization, Writing – original draft,
529 Writing – review & editing. **Richar Andre Vilca-Solorzano:** Conceptual-
530 ization, Data curation, Formal analysis, Investigation, Methodology, Project
531 administration, Resources, Software, Validation, Visualization, Writing –
532 original draft, Writing – review & editing. **Fred Torres-Cruz:** Conceptual-
533 ization, Data curation, Formal analysis, Investigation, Methodology, Project
534 administration, Resources, Software, Supervision, Validation, Visualization,
535 Writing – original draft, Writing – review & editing. **Vladimiro Ibañez:**
536 Conceptualization, Data curation, Formal analysis, Investigation, Methodol-
537 ogy, Project administration, Resources, Supervision, Validation, Visualiza-
538 tion, Writing – original draft, Writing – review & editing.
539 All four authors contributed substantively across every stage of the study,
540 and all read and approved the final manuscript.

541 **Declaration of competing interest**

542 The authors declare that they have no known competing financial inter-
543 ests or personal relationships that could have appeared to influence the work
544 reported in this paper.

545 **Data availability**

The satellite–field match-up datasets and the complete analysis pipeline, including the scripts that generate every number, table and figure in this article, are openly available at the repository below. Raw imagery is freely available via Google Earth Engine; the in-situ measurements derive from published IMARPE/ALT and OEFA monitoring records.
547 <https://github.com/Andre031222/harmonized-titicaca-transparency>



Repository

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